

Lab 3 Report – Social Robotics – Leali Marco – 07/10/2025

Part A

In this lab, I implemented new gestures and user interaction features for Furhat. I created three custom gestures: **BigSmile**, **NeckMovement** and **EyeRoll**. Furhat can track users in the environment; it attends to the closest detected user or the front location if no users are detected. These gestures were integrated into a short demonstration dialogue where Furhat greets the user with “Hi” and asks “How are you?”. FurHat responds with appropriate gestures depending on the user’s input. Furthermore, I implemented a gesture with audio, **SurprisedWithSound**, where Furhat performs a surprised facial expression accompanied by an elephant sound, if the user says a phrase in which is included the word “wow”. If the word included is “bad”, an extract from an Eminem’s song starts. It is a loop conversation until the user doesn’t include in the utterance the word “bye”. Furhat will say: “It was nice talking to you! Goodbye”

Part B

Furhat was integrated with a **LLM-based dialogue system** (continuing from Lab 2). User speech is captured, analyzed and used as input to the LLM (llama3.1), which generates contextually relevant responses. Furhat responds using speech and can trigger additional gestures based on the user’s input or LLM content. For example, if the user says “wow,” Furhat replies and performs a surprise gesture with sound; if the user says “bad,” it apologizes, plays audio and smiles. This integration demonstrates Furhat acting as a conversational interface powered by a large language model, combining verbal and non-verbal communication.

Conclusion and next Improvements

The lab demonstrates how gestures, user tracking and external LLM dialogue systems can be combined to create engaging, multimodal interactions with Furhat. This setup provides a foundation for more complex social robotic interactions. This lab was just a short demonstration of the potential of FurHat. The conversation implemented in Part A is very trivial and it could be improved a lot. The LLM-based dialogue system in Part B is better, but it must be improved as well. Gestures and sounds can be added, but, most of all, the model itself can be improved adding control parameters for the fluency of the speech and for a better human-robot interaction.